Behavioural adaptations to environment

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The environment a species inhabits will affect how it adapts and evolves. Dispersion, or 'spreading-out' activity, plays an important role in many behavioural adaptations. Dispersion (*disperse* to part) can involve animals or their food supply. For example, an animal that must travel long distances at regular intervals in order to eat will need, among other characteristics, the strength and stamina to withstand the trips. Natural selection will favour individuals who possess these features.

Other ecological adaptations are associated with daily life. Flexibility is important. Some important factors to consider include:

- Is it more advantageous for the species to live in groups or alone?
- If the species lives in groups, is it better to work together as a co-operating community (eg ants, bees), or to just live as individuals who happen to share a habitat? This point applies to territory as well. Should the animals defend their whole territory as a group or just defend their own small portion of it from others (whether they are members of the community or not).
- Should parental care be long-term or non-existent? For long-term care, would it be more beneficial for one or both parents to care for their own offspring or, as in the case of lionesses, the whole group protects the young, with several females feeding cubs whenever possible.

1 *The ability to adapt one's behaviour helps to maximise an organism's fitness.* Discuss this statement.

2 Describe the difference between *adapted* and *inherited*.

3 Suggest ways in which available foods can lead to behavioural adaptations.

4 Suggest ways in which available foods can lead to evolution of a trait.

The tiger salamander (*Ambystoma tigrinum*) has a cannibalistic form, or *morph*. When its aquatic environment becomes overcrowded and food shortages occur, certain of the larger individuals turn cannibalistic, eating smaller members of their own species.

5 Suggest how these morphs might maximise the survival of the species within the pond.

6 Discuss how the different types of social organisation might affect how a species adapts.

7 Discuss two advantages and two disadvantages of the different types of parental care.

Identify the terms defined by the following statements:

8 *Organism that eats it own kind.*

9 *Place where an organism usually lives.*

10 *Process of gradual change to a better state.*

Jean Lamarck

Many people have noticed that animals change over time and have come up with explanations for these changes. Jean Lamarck (1744–1829) proposed that if an animal changed during its lifetime these alterations were passed on to the offspring. The most quoted example of his theory is that of how giraffes gained their long necks.

Giraffes have long necks especially suited to grazing on tall trees. According to the Lamarck theory, giraffes developed long necks by a gradual process. In times of plenty, all giraffes ate well. When drought came and forage was in short supply, giraffes stretched to eat any food that was available and gained long necks through the stretching. Those giraffes with long necks survived and passed on their characteristics to the next generation, who continued to stretch for food and get longer necks. Over many generations giraffes developed long necks.

11 Most dogs have tails. Some dog breeders dock (cut off tails) young dogs because the particular breed might be more fashionable without a tail. Explain why no matter how many tails of young dogs are removed, the offspring will still have tails.

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